

ard_k210-Color recognition

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1.K210 and Arduino communication

1.1 Experimental premises

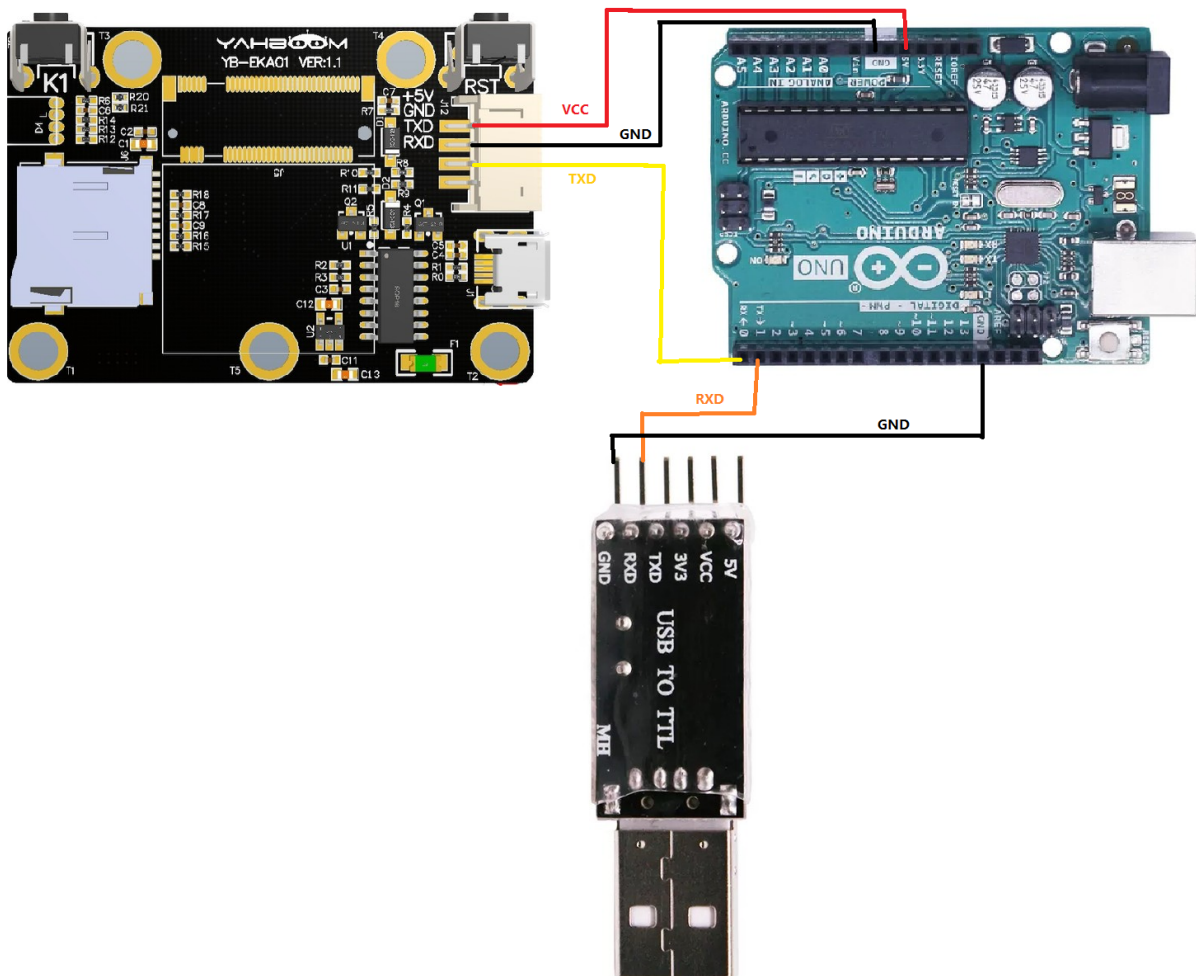
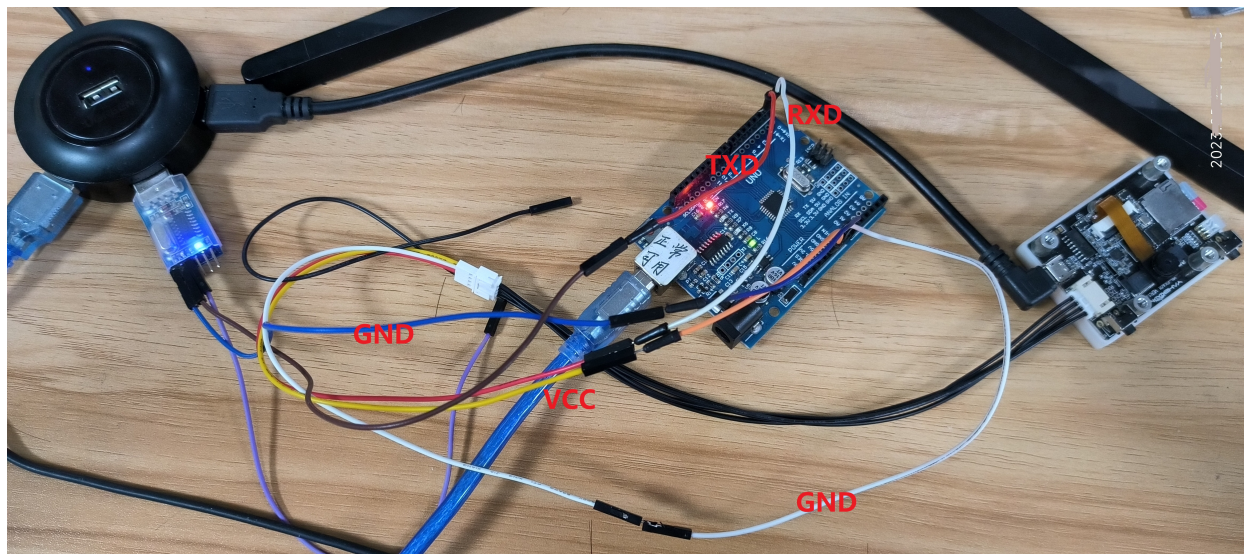
This tutorial uses arduino, and K210 requires running the program in **K210-AI (stm32_pico_arduino)** to start the experiment

- arduino *1
- K210 perspective module * 1 (requires SD card (with AI model inside) and camera)
- USB to TTL module * 1

1.2 Experimental wiring

arduino	usb to ttl
TXD	RXD
GND	GND

arduino	K210 perspective module
RXD	TXD
GND	GND
VCC	5V
Wiring as shown in the diagram :	



1.3 Main code explanation

```
void loop()
{
    while (K210Serial.available())
    {
```

```

recv_k210msg(k210Serial.read());

if (k210_msg.class_n != 0)
{
    if(k210_msg.class_n == 1)
    {

        sprintf(buff_com, "x=%d,y=%d,w=%d,h=%d\r\n", k210_msg.x, k210_msg.y, k210_msg.w, k210_msg.h);

        k210Serial.print(buff_com);

        k210_msg.class_n = 0;
    }

}

}
}

```

After the above program, if you are running this routine, k210_ The members of the msg structure have corresponding values and are processed through serial port printing

K210_Msg: is a structure that receives information, and its main members are

- X: is the horizontal coordinate of the top left corner of the recognized box (range: 0-240)
 - Y: is the vertical coordinate of the upper left corner of the identified box (range: 0-320)
 - W: is the width of the recognized box (range: 0-240)
 - H: The length of the recognized box (range: 0-320)
 - ID: is the recognized label
 - Class_n: Routine number
 - Msg_Msg [20]: Valid data
- After receiving and processing data, k210_ Each member of the msg will store valid information. If you want to develop it again, call K210 directly_ Members of msg are sufficient

1.4 experimental phenomena

1. After connecting the cable, the K210 perspective module runs offline. Please check 【6.2 K210 as coprocessor】 -- 【ReadMe】

2. Set the serial port assistant to the interface shown in the figure

COM Configs

Channel: COM1 # 前

Baudrate: 115200

Paritybits: NONE

Databits: 8

Stopbits: 1

Flowctrl: NONE

Close

Recv Options

☒ ASCII ☐ HEX

☒ Log display mode

☒ Auto linefeed

☐ Hide received data

☐ Recv save to file...

[AutoScroll](#) [Clear](#)

Send Options

☒ ASCII ☐ HEX

☒ Enable escape chars

☒ AT CMD auto CRLF

☐ Auto append bytes

☐ Send from file ...

☐ Period: 1000 ms

[Shortcut](#) [History](#)

3. Then run the color recognition routine, and the serial assistant will print out the important information transmitted from k210 to arduino, as shown in the following figure

```
x=0, y=0, w=319, h=106
x=0, y=0, w=319, h=95
x=0, y=0, w=319, h=85
x=0, y=0, w=319, h=73
x=177, y=0, w=142, h=62
x=192, y=0, w=127, h=55
x=197, y=0, w=122, h=52
x=202, y=0, w=117, h=50
x=204, y=0, w=115, h=48
x=201, y=0, w=118, h=49
x=202, y=0, w=117, h=49
x=206, y=0, w=113, h=47
x=209, y=0, w=110, h=47
x=209, y=0, w=110, h=47
x=209, y=0, w=110, h=47
x=211, y=0, w=108, h=46
x=215, y=0, w=104, h=44
x=216, y=0, w=103, h=44
x=216, y=0, w=103, h=43
x=218, y=0, w=101, h=43
x=219, y=0, w=100, h=42
x=219, y=0, w=100, h=42
x=218, y=0, w=101, h=41
x=219, y=0, w=100, h=40
x=224, y=0, w=95, h=36
```

Color recognition only transmits the four Member variable of k210_msg: x, y, w, h.

